

NIRIKSHAK (निरीक्षक) — 5-Minute Compact Master Pitch Script (Hinglish)

Smart India Hackathon (SIH26034) | Ministry of Consumer Affairs, Food & Public Distribution

Department of Consumer Affairs (DoCA), Government of India

1st-Principles Compact Strategy

- **Target Duration:** Exactly **5 Minutes** (300 Seconds) with breathing room for pauses.
- **Spoken Word Budget:** **~520–560 Words** (Paced at ~110–120 WPM — slow, confident, perfectly articulated).
- **Core Signal:** Har line mein sirf **verified facts, exact law, aur working engineering math** hai — zero fluff, zero hype.
- **Delivery Mode:** Har major section mein **[OPTION A]** (Dynamic / High Energy) aur **[OPTION B]** (Authoritative / Calm Gravitas) diya gaya hai.

5-MINUTE TIMING & PACING BUDGET			
SECTION	DURATION	WORD COUNT	CORE THEME
PART 1: The Opening Rule	0:00 - 1:00 (60s)	~110 words	Deficit & Golden

PART 2: Core Engineering Rules	1:00 - 3:00 (120s)	~230 words	CV, Homography &
PART 3: Viability & Law Vishwas	3:00 - 4:40 (100s)	~190 words	4 SKUs & Jan
Closing Punchline	4:40 - 5:00 (20s)	~40 words	Impact & Handoff
TOTAL Noise	5 Minutes (300s)	~570 words	100% Signal / 0%



THE COMPACT 5-MINUTE PITCH

PART 1: THE OPENING (The National Deficit & The NIRIKSHAK Thesis)

0:00 – 1:00 (60 Seconds) | Budget: ~110 Words

(Stage Cue: Speaker stands tall, maintains eye contact, speaks with steady conviction)

[0:00 - 0:30] The 1:4,000 Manpower Crisis

[OPTION A - Scale & Ground Reality]:

"Respected Judges, India mein har saal ₹25 Lakh Crore se zyada ke packaged consumer goods bikte hain — 1.2 Crore retail shops aur wholesale mandis ke through.

Lekin pure desh mein compliance inspect karne ke liye hamare paas sirf ~3,000 Legal Metrology Officers hain!

Yani ek officer par 4,000 se zyada establishments!

Result? 99% se zyada retail packaging bina kisi inspection ke market mein circulate hoti hai."

[OR OPTION B - Citizen Trust & Enforcement Gap]:

"Respected Judges, subah ke biscuit se lekar life-saving dawaaiyon tak — 140 Crore nagrik rozana pre-packaged products par trust karte hain.

Lekin 1.2 Crore retail dukanon ke samne pure desh mein sirf ~3,000 Legal

Metrology Officers hain. Ek inspector ke hisse 4,000 establishments aate hain — jisse manual inspection physically impossible ho chuki hai."

[0:30 - 1:00] Manual Bottleneck & NIRIKSHAK Golden Rule

"Kyunki ek multi-panel carton ko vernier caliper aur magnifying glass se inspect karna **20 se 25 minute** leta hai. Aur 1 July 2024 ke baad, **Section 63 of Bharatiya Sakshya Adhiniyam (BSA) 2023** ke tehat unverified smartphone photos court mein sidhe challenge ho jaati hain.

We present **NIRIKSHAK** — an AI-Assisted Legal Metrology Workstation. Hamara operating thesis non-negotiable hai:

'AI Observes, Deterministic Rules Verify, Human Officer Decides.'

System 25-minute ke manual audit ko **under 2 minutes** mein compress karta hai."

PART 2: THE CORE TECH (Calibration Math, Vision & Deterministic AST)

 1:00 – 3:00 (120 Seconds) | Budget: ~230 Words

(Stage Cue: Point to Pipeline architecture slide)

[1:00 - 1:30] Why Zero Generative AI in Legal Decisions?

"Sabse pehle: **'Kya AI hallucinate karke galat notice bhejega?'**

Hamara answer hai: **Bilkul nahi!**

Generative LLMs probabilistic hote hain — unpar legal penalties decide nahi ho sakti.

NIRIKSHAK mein Deep Learning sirf character aur polygon detection tak simit hai. Legal compliance hamara **Deterministic AST Rule Engine** karta hai — exact Gazette clauses, zero hallucination!"

[1:30 - 2:15] Innovations 1 & 2: Planar Homography & Multilingual Edge Vision

(Stage Cue: Show reference card / calibration visual)

[OPTION A - Direct Math & Caliper Precision]:

*"Pehli Innovation: Planar Homography Calibration.

2D photo mein pixels hote hain, millimeters nahi. Inspector product ke sath standard ISO-7810 card ya ArUco marker rakhta hai. Hamara system **3x3 Homography Matrix (H)** se perspective tilt un-tilt karta hai aur pixels ko exact real-world millimeters mein convert karta hai.

Real physical test mein digital caliper reading **1.47 mm** thi — Nirikshak ne measure kiya **1.46 mm** — delta sirf **0.01 mm**!

Doosri Innovation: Multilingual Edge Vision.

DBNet++ aur PP-OCRv4 INT8 quantized ONNX models standard CPU par English aur Devanagari Hindi ko sirf **180 milliseconds** mein padhte hain — with **₹0 Cloud GPU expense**."

[OR OPTION B - Algorithmic Rigor]:

"Linear pixel scaling tilt hone par fail ho jaati hai. Hum 4 coplanar points se Planar Homography calculate karke perspective skew ko geometrically rectify karte hain — achieving sub-0.30mm verified accuracy.

Multilingual OCR ke liye hum DBNet++ aur PP-OCRv4 CPU-quantized pipeline use karte hain, jo Latin aur Devanagari Hindi dono scripts ko 180ms mein local hardware par transcribe karti hai."

[2:15 - 3:00] Innovations 3 & 4: 3D Semantic Fusion & Section 63 BSA Vault

(Stage Cue: Point to 3D multi-panel package and PDF certificate)

*"Teesri Innovation: Cross-Facet Semantic Fusion.

Packages 3D hote hain — brand aage hai, MRP side mein, consumer care peeche. Inspector multi-angle photos leta hai, aur Fusion Engine spatial graph se saare tokens ko link karke composite legal docket assemble karta hai.

Chauthi Innovation: Section 63 BSA 2023 Evidentiary Vault.

Har raw camera buffer, GPS coordinate, aur device fingerprint ka **SHA-256**

Merkle DAG Root banta hai, jo official Form-1 Notice PDF mein digitally seal hota hai. Image ka ek single pixel bhi tamper hone par cryptographic certificate reject ho jayega. Courtroom mein evidence undeniable hai!"*

PART 3: FEASIBILITY, VIABILITY & NATIONAL IMPACT

🕒 3:00 – 4:40 (100 Seconds) | Budget: ~190 Words

(Stage Cue: Direct judges to 4-SKU empirical table)

[3:00 - 3:45] 4 Real Retail SKUs & Jan Vishwas Proportional Justice

*"Humne synthetic data par nahi, market se kharide **4 real physical products (38 photos)** par system validate kiya:

1. **Titan Watch Box:** Clean PASS. Single-unit package hone se Rule 6(1)(da) ke tehat Unit Sale Price exempt tha — system ne exemption correctly pehchani.
2. **Himalaya Brahmi:** FAIL on technical deficit. Batch stamp font 1.46mm nikla bajaye 2.0mm ke.
3. **Boult Audio Earbuds:** FAIL on thermal sticker font deficit (1.24mm vs 2.0mm).
4. **Gopi Baba Hair Oil:** FAIL on 3 substantive violations: non-standard unit **100 ml.** with dot, missing tax clause, aur missing consumer email — ₹25,000 Compounding Notice!

Dhyan dijiye: **Jan Vishwas Act, 2023** ke mutabiq packaging offenses decriminalized hain.

Himalaya jaise minor printing defect par system **15-Day Improvement Notice with ₹0 penalty** deta hai taaki business agle batch mein sudhaar sake — true Ease of Doing Business!"*

[3:45 - 4:40] Operational Viability & Scalability

[OPTION A - 13x Throughput & Fiscal Viability]:

*"Government exchequer ke liye viability teen baaton par grounded hai:

1. **13x Throughput:** Per-package time 20 minute se ghata kar under 2 minute hota hai — officer capacity **15 se badh kar 200+ packages daily** ho jaati hai.
2. **₹0 Cloud Cost:** CPU inference aur Mode B (Offline SQLite) ki wajah se rural mandis mein zero internet par chalta hai.
3. **DoCA Integration:** National eMaap portal ke sath standardized JSON schema sync already architected hai."*

[OR OPTION B - Ecosystem Protection]:

"NIRIKSHAK pure packaging ecosystem ko protect karta hai:

Field inspectors ko automated drafting milti hai,

*Manufacturers **Pre-Compliance Simulator** se karodon ke product recall se bachte hain,*

Aur Courts ko Section 63 BSA ke tehat unshakeable digital evidence milta hai."

CLOSING PUNCHLINE & DEMO HANDOFF

 **4:40 – 5:00 (20 Seconds) | Budget: ~40 Words**

(Stage Cue: Speaker looks directly at panel, calm and confident)

[OPTION A - Sovereign Authority Closing]:

*"NIRIKSHAK legal metrology enforcement ko manual caliper disputes se nikaal kar **calibrated precision, transparent adjudication, aur digital sovereignty** deta hai.*

'AI Observes, Deterministic Rules Verify, Officer Decides.'

Thank you, Respected Judges! We invite you to our live prototype demonstration."

[OR OPTION B - Patriotic Mission Closing]:

"NIRIKSHAK protects 140 Crore Indian consumers, empowers our enforcement officers, and safeguards honest businesses.

Thank you. We are now eager to take your questions and demonstrate the live system!"



30-SECOND ELEVATOR PITCH (In Case Judges Say: "Give me the summary in 30 seconds!")

"Sir, India mein 1.2 Crore retail shops ke samne sirf 3,000 Legal Metrology Officers hain, aur manual inspection mein 20-25 minute lagte hain. NIRIKSHAK ek AI-assisted inspection workstation hai jo Planar Homography se phone camera ko sub-millimeter caliper banata hai (0.01mm verified accuracy). Deep learning sirf OCR karta hai, jabki legal compliance 100% deterministic code execute karta hai. Under Jan Vishwas Act 2023 aur Section 63 BSA 2023, yeh 2 minute mein tamper-proof electronic notice generate karta hai — with ₹0 Cloud GPU cost!"

Compact Edition · Optimized for Maximum Stage Signal & Precision Delivery